14.05.2025

256 MiB MM

32 KiB cache 1 word = 32 bits = 4B

byte - addressable

i) DM, 4 words / block, write-through

ii) 4 way SA / 8 words / block, write-back, LRU → least recent used
↳ la replace ne uităm care info a fost folosită ultima

a) index, tag, offset size

b) total cache size

c) starting with power on, following byte-addressed references are made: 0x00, 0x40, 0x8C, 0xBA, 0x400, 0xC1C

index, tag, valid

d) hit ratio

i) $256\ MiB^{MM} = 2^8 \cdot 2^{20} = 2^{28} B \Rightarrow P.A. = 28$ bits

1 word = 32 bits = 4 B = 2^2 B ⇒ 2 word offset

32 KiB cache = $2^5 \cdot 2^{10} = 2^{15}$ B

4 words / block ⇒ 2^{2} → block offset

$\frac{2^{15}}{2^4} = 2^{11}$ blocks ⇒ index = 11

27 tag	index	B.O.	W.O.
<13>	<11>	<2>	<2>

b) $2^{11}(1 + 13 + \underbrace{2^7}_{2^2 \cdot 2^2 \cdot 8}) = 2^{11} \cdot 142$

c)

	tag	index	V	offset	H/M	replaced
0x00 =	0	0	0	0	M	NO
0x40 =	000...	0100	0	0000	M	NO
0x8C =	000...	1000	0	1100	M	NO
0xBA =	0 ...	1011	0	1010	M	NO

0x400	= 0100	0000	1	0000	M	0x00 - 0x10
0xC1C	1100	0001	0	1100	M	NO
0xC11	1100	0001	1	0001	H	NO

d) Hit Ratio $= \frac{1}{7} = 14,2\%$

ii) a) 8 words/block $\Rightarrow 2^3 \Rightarrow$ 3 block offset

256 MiB $= 2^8 \cdot 2^{20} = 2^{28}$ B $\Rightarrow$ P.A. = 28 bits

1 word = 32 bits = 4 B $= 2^2$ B $\Rightarrow$ 2 word offset

32 KiB cache $= 2^5 \cdot 2^{10} = 2^{15}$ B

$\frac{2^{15}}{2^3 \cdot 2^2} = 2^{10} \Rightarrow$ index $= \frac{2^{10}}{2^2} = 2^8 \Rightarrow$ index = 8

$\rightarrow$ 4 way SA

tag = 28 - 8 - 5 = 15

b) $2^{10}(1 + 15 + 1 + \underset{\text{4 way}}{2} + \underset{\text{B/w}}{2^2} \cdot \underset{\text{w/Block}}{2^3} \cdot 2^3)$ bits =

$= 2^{10}(19 + 2^8) = 275 \cdot 2^{10}$

L_{256}

total cache size $= 2^8 \cdot 2^2 \cdot (\underset{\text{valid}}{1} + \underset{\text{tag bits}}{15} + \underset{\text{dirty bit}}{1} + \underset{\text{2 bits age}}{2} + 2^2 \cdot 2^3 \cdot 2^3)$ bits

2^8 intrări 4 cache-uri

4 way

$= 275 \cdot 2^{10}$

c)

	tag	index	V	offset	H/M	replaced
0x00 →	0	0	0	00000	M	NO
0x40 →	0	..0010	0	00000	M	NO
0x8C →	0	..0100	0	01100	M	NO
0xBA →	0	...0101	0	11010	M	NO
0x400 →	0	100000	0	00000	M	NO
0xC1C →	0	1100000	0	11100	M	NO
0xC11	0	1100000	1	10001	H	NO
cerva en tag	1	1100000	0	10001	M	NO (am mers în alt cache)

Cache performance

$CPU_{TIME} = IC \times CPI \times CCT$
↳ afectat de memorie

CPi → toate ref de memorie sunt hit

$CPI_{real} = CPI_{ideal}$ + mem access stall cycles

mem access stall cycles = MM access time × MR × $\frac{\text{memory accesses}}{\text{instr}}$

miss penalty (MP)

= MM accesses × MR × MP

ex:

$M_{R_{IC}} = 2\%$

$M_{R_{DC}} = 4\%$

main memory access time 100 CC

36% of all instr are L/S

CPi ideal = 2

a) How much faster would this

$$\frac{CPI_{real}}{CPI_{ideal}} = \frac{2 + \overbrace{1 \times 0,02 \times 100}^{\text{i.c.}} + \overbrace{0,004 \times 0,36 \times 100}^{\text{D.C. accesses}}}{2} =$$

$$= \frac{2 + 2 + 1,44}{2} = 2,72$$

b) −||− $= \frac{1 + 2 + 1,44}{1} = \frac{4,44}{1} = 4,44$

AMAT = average memory access time

AMAT = access / hit time + MR × MP [s]

1. $CPI_{ideal} = 1$

CR = 4 GHz